

MTH-100

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Question No 1

Let $F(x) = \frac{2x^2 - 5x}{x^2 - 3}$ and $G(x) = \sqrt{19 - x^2}$

a) Determine The domain and range of each functions :

Domain:

To find the domain of $f(x)$ we need to ensure that the denominator $x^2 - 3$ is not zero, as division by zero is undefined.

1. Set the denominator equal to zero:

$$x^2 - 3$$

$$x^2 = 3$$

$$x = \pm\sqrt{3}$$

2. So, $f(x)$ is undefined at $x = \sqrt{3}$ and $x = -\sqrt{3}$

• **Domain:**

$$(-\infty, -\sqrt{3}) \cup (-\sqrt{3}, \sqrt{3}) \cup (\sqrt{3}, \infty)$$

Range:

Since this is a rational function with a cubic polynomial in the numerator and a quadratic polynomial in the denominator, it will approach positive or negative infinity as $x \rightarrow -\infty$ and $x \rightarrow \infty$. However, determining the exact range analytically is complex and typically requires calculus (finding asymptotes, critical points, etc.). Therefore, we can say:

- Range: $(-\infty, \infty)$ except possibly at values excluded by vertical asymptotes or points of discontinuity.

2. Function : $G(x) = \sqrt{19 - x^2}$

Domain:

For $g(x)$ to be defined, the expression inside the square root must be non-negative:

1. Set the inside of the square root to be greater than or equal to zero:

$$19 - x^2 \geq 0$$

$$x^2 \leq 19$$

$$-\sqrt{19} \leq x \leq \sqrt{19}$$

2. So, $g(x)$ is defined for x in the interval $[-\sqrt{19}, \sqrt{19}]$

- Domain: $[-\sqrt{19}, \sqrt{19}]$

Range:

Since $g(x) = \sqrt{19 - x^2}$ represents the upper half of a circle with radius $\sqrt{19}$, the values of $g(x)$ range from 0 to $\sqrt{19}$

Range: $(0, \sqrt{19})$

Summary

- For $f(x) = \frac{2x^3 - 5x}{x^2 - 3}$
- Domain: $(-\infty, -\sqrt{3}) \cup (-\sqrt{3}, \sqrt{3}) \cup (\sqrt{3}, \infty)$
- Range: $(-\infty, \infty)$ (except possibly at points of discontinuity)
- For $g(x) = \sqrt{19 - x^2}$
- Domain: $(-\sqrt{19}, \sqrt{19})$
- Range: $(0, \sqrt{19})$

b) Classify each function as even odd or neither justify your answer with the necessary algebraic Checks

To determine whether each function is even, odd, or neither, we need to apply the definitions of even and odd functions.

- A function $f(x)$ is even if $f(-x) = f(x)$ for all x in its domain.
- A function $f(x)$ is odd if $f(-x) = -f(x)$ for all x in its domain.
- If neither of these conditions is satisfied, the function is classified as neither even nor odd.

Let's check each function.

1. Function $F(x)$

To determine if $f(x)$ is even, odd, or neither, let's calculate $f(-x)$

1. Substitute $-x$ for x in $f(x)$:

$$F(-x) = \frac{2(-x^3) - 5(-x)}{(-x)^2 - 3}$$

2. Simplify $f(-x)$

$$F(-x) = \frac{2(-x^3) + 5x}{x^2 - 3}$$

$$= \frac{-2x^3 + 5x}{x^2 - 3}$$

3. Compare $f(x)$ to $f(-x)$:

- $f(x) \neq f(-x)$
- $f(x) \neq -f(-x)$

Since $f(-x) \neq f(x)$ and $f(-x) \neq -f(x)$, the function $f(x)$ is neither even nor odd.

2. Function $g(x)$

To determine if $g(x)$ is even, odd, or neither, let's calculate $g(-x)$.

1. Substitute $-x$ for x in $g(x)$:

$$G(-x) = \sqrt{19 - (-x)^2}$$

2. Simplify $g(-x)$

$$G(-x) = \sqrt{19 - x^2}$$

3. Compare $g(-x)$ to $g(x)$:

- $g(x) = \sqrt{19 - x^2}$
- $g(-x) = \sqrt{19 - x^2}$

Since $g(-x) = g(x)$ for all x in the domain, $g(x)$ is even.

Summary

- For $f(x)$: The function is neither even nor odd.
- For $g(x)$: The function is even.

C) For $f(g(x))$ and $g(f(z))$ Find the domains and explain any restrictions

To find the domains of $f(g(x))$ and $g(f(z))$, we need to understand the compositions of the functions $f(x)$ and $g(x)$, and determine any restrictions that arise in these compositions.

1. Finding the Domain of $f(g(x))$

To form $f(g(x))$, we need to substitute $g(x)$ into $f(x)$. The domain of $f(g(x))$ will depend on both:

- The domain of $g(x)$, since $g(x)$ must be defined.

- The domain of $f(x)$ evaluated at $g(x)$, since $f(g(x))$ must be defined.

Step 1: Domain of $g(x) = \sqrt{19 - x^2}$

As previously calculated:

- The domain of $g(x)$ is $x \in (-\sqrt{19}, \sqrt{19})$ since $19 - x^2 \geq 0$

Step 2: Substitute $g(x)$ into $f(x)$ to form $f(g(x))$

Since $f(x) = \frac{2x^3 - 5x}{x^2 - 3}$ substituting $G(x) = \sqrt{19 - x^2}$ into $f(x)$ gives:

$$F(g(x)) = f(\sqrt{19 - x^2}) = \frac{2(\sqrt{19 - x^2})^3 - 5(\sqrt{19 - x^2})}{(\sqrt{19 - x^2})^2 - 3}$$

Step 3: Determine Additional Restrictions from $\frac{2(\sqrt{19 - x^2})^3 - 5(\sqrt{19 - x^2})}{(\sqrt{19 - x^2})^2 - 3}$

To find additional restrictions, we need to ensure that the denominator in $f(g(x))$ does not equal zero:

$$(\sqrt{19 - x^2})^2 - 3 \neq 0$$

$$19 - x^2 - 3 \neq 0$$

$$16 - x^2 \neq 0$$

$$x^2 \neq 16$$

$$x \neq \pm 4$$

Thus, $x = \pm 4$ are values that would make the denominator zero, so they are excluded from the domain of $f(g(x))$

Domain of $f(g(x))$

Combining the restrictions from the domain of $g(x)$ and the restriction $x \neq \pm 4$, we get:

- Domain of $f(g(x))$

$$x \in (-\sqrt{19}, \sqrt{19}) \setminus (-4, 4)$$

2. Finding the Domain of $g(f(z))$

To form $g(f(z))$ we need to substitute $f(z)$ into $g(x)$. The domain of $g(f(z))$ will depend on both:

- The domain of $f(z)$, since $f(z)$ must be defined.
- The domain of $g(x)$ evaluated at $f(z)$, since $g(f(z))$ must be defined.

Step 1: Domain of $F(z) = \frac{2z^3 - 5z}{z^2 - 3}$

As previously calculated:

- The domain of $f(z)$ is $z \in (-\infty, -\sqrt{3}) \cup (-\sqrt{3}, \sqrt{3}) \cup (\sqrt{3}, \infty)$, since $z = \pm\sqrt{3}$ would make the denominator zero.

Step 2: Substitute $f(z)$ into $g(x)$ to form $g(f(z))$

Since $g(x) = \sqrt{19 - x^2}$, substituting $f(z)$ into $g(x)$ gives:

$$G(f(z)) = g\left(\frac{2z^3 - 5z}{z^2 - 3}\right) = \sqrt{19 - \left(\frac{2z^3 - 5z}{z^2 - 3}\right)^2}$$

Step 3: Determine Additional Restrictions from $g(f(z))$

To ensure $g(f(z))$ is defined, the expression inside the square root must be non-negative:

$$19 - \left(\frac{2z^3 - 5z}{z^2 - 3}\right)^2 \geq 0$$

This inequality is complex to solve algebraically, but a key observation is that we need:

$$\left| \frac{2z^3 - 5z}{z^2 - 3} \right| \leq \sqrt{19}$$

Furthermore, $f(z)$ must be in the domain of $g(x)$ which is $(-\sqrt{19}, \sqrt{19})$. This adds a restriction to the values of z , limiting $f(z)$ within that range. A precise determination of this subset of z -values would typically require numerical or graphical analysis.

Domain of $g(f(z))$

In general terms, the domain of $g(f(z))$ includes values of z in the domain of $f(z)$ such that $f(z)$ is in the domain of $g(x)$ and excludes points where the expression inside the square root becomes negative or where $f(z)$ itself is undefined.

Question No 2

Solve the following system of equations using the reduced row echelon form (RREF) method:

$$\begin{aligned}2x + 3y - z &= 5 \\ 4x + y + 2z &= 6 \\ +5y + 3z &= -4\end{aligned}$$

a) Set up an augmented matrix for this system and perform row operations to achieve the RREF.

To solve the given system of equations using the reduced row echelon form (RREF) method, let's first write the system properly and form the augmented matrix.

Let's represent this system as an augmented matrix:

$$\left(\begin{array}{ccc|c} 2 & 3 & -1 & 5 \\ 4 & 1 & 2 & 6 \\ 0 & 5 & 3 & -4 \end{array} \right)$$

Now, we'll use row operations to convert this matrix to RREF.

Step 1: Make the Leading Entry in Row 1 Equal to 1

Divide Row 1 by 2:

$$\left(\begin{array}{ccc|c} 1 & \frac{3}{2} & -\frac{1}{2} & \frac{5}{2} \\ 0 & -5 & 4 & 6 \\ 0 & 5 & 3 & -4 \end{array} \right)$$

Step 2: Make the Entry Below the Leading 1 in Column 1 Equal to 0

Subtract 4 times Row 1 from Row 2:

$$\left(\begin{array}{ccc|c} 1 & \frac{3}{2} & -\frac{1}{2} & \frac{5}{2} \\ 0 & -5 & 4 & 6 \\ 0 & 5 & 3 & -4 \end{array} \right)$$

Step 3: Make the Entry Below the Leading Entry in Row 2 Equal to 0

Add Row 2 to Row 3:

$$\left(\begin{array}{ccc|c} 1 & \frac{3}{2} & -\frac{1}{2} & \frac{5}{2} \\ 0 & -5 & 4 & -4 \\ 0 & 5 & 3 & -4 \end{array} \right)$$

Step 4: Make the Leading Entry in Row 3 Equal to 1

Divide Row 3 by 7:

$$\left(\begin{array}{ccc|c} 1 & \frac{3}{2} & -\frac{1}{2} & \frac{5}{2} \\ 0 & -5 & 4 & -4 \\ 0 & 0 & 7 & -\frac{8}{7} \end{array} \right)$$

Step 5: Make the Entries Above the Leading 1 in Column 3 Equal to 0

1. Add $\frac{1}{2}$ times Row 3 to Row 1:

$$\left(\begin{array}{ccc|c} 1 & \frac{3}{2} & 0 & \frac{11}{4} \\ 0 & -5 & 4 & -4 \\ 0 & 0 & 1 & -\frac{8}{7} \end{array} \right)$$

2. Subtract 4 times Row 3 from Row 2:

$$\left(\begin{array}{ccc|c} 1 & \frac{3}{2} & 0 & \frac{11}{4} \\ 0 & -5 & 0 & \frac{8}{7} \\ 0 & 0 & 1 & -\frac{8}{7} \end{array} \right)$$

Step 6: Make the Leading Entry in Row 2 Equal to 1

Divide Row 2 by -5:

$$\left(\begin{array}{ccc|c} 1 & \frac{3}{2} & 0 & \frac{11}{4} \\ 0 & 1 & 0 & -\frac{8}{35} \\ 0 & 0 & 1 & -\frac{8}{7} \end{array} \right)$$

Step 7: Make the Entry Above the Leading 1 in Column 2 Equal to 0

Subtract $\frac{3}{2}$ times Row 2 from Row 1:

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & \frac{9}{5} \\ 0 & 1 & 0 & -\frac{8}{35} \\ 0 & 0 & 1 & -\frac{8}{7} \end{array} \right)$$

Solution

Now that we are in reduced row echelon form, we can read off the solutions:

$$X = \frac{9}{5}, y = -\frac{8}{35}, Z = -\frac{8}{7}$$

b. Interpret the solution. Is it unique, or does it represent a family of solutions? If it represent* family, describe the solution set in terms of free variables.

The solution to the given system of equations is unique. Here's why:

In an augmented matrix, a unique solution occurs when each variable corresponds to a pivot column (a column with a leading 1) and there are no free variables. If we look at the final reduced row echelon form (RREF) matrix:

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & \frac{9}{5} \\ 0 & 1 & 0 & -\frac{8}{35} \\ 0 & 0 & 1 & -\frac{8}{7} \end{array} \right)$$

Each variable x, y, and z is associated with a pivot column, and no columns are left without a leading 1 in any of the rows (no free variables). Therefore, each variable has a single, specific value:

$$x = \frac{9}{5}, y = -\frac{8}{35}, z = -\frac{8}{7}$$

Since all variables are determined and there are no free variables, this solution represents a single point in three-dimensional space rather than a family of solutions.

Thus, the solution is unique and does not represent a family of solutions.